

Measurements of the loss probability of nitrogen atoms versus temperature on various surfaces

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Abstract

The loss probability γ_N of nitrogen atoms is measured versus the temperature on different surfaces by a method based on optical emission spectroscopy measurements. In a cylindrical quartz tube, a late Ar-N₂ post-discharge is used. Emission of the first positive system is observed in a zone that precedes a heated zone and in another zone that follows it. Each observed zone is at room temperature. Indeed, in these conditions, nitrogen atoms titration by NO permits the attainment of the absolute value of the N density. This method can give the dependence of the loss probability of nitrogen atoms versus the temperature. The very simple geometry used here enables the use of foils made of any material introduced in the heated zone of the reactor. Therefore, different wall materials can be easily studied. Calculation of the loss probability γ_N requires a 2D hydrodynamic, thermal and kinetics modelling. It was developed and it included, diffusion to the walls, temperature gradients, volume and surface nitrogen atom consumptions. The first results obtained on quartz show that N atoms recombine according to an Eley–Rideal mechanism and $\gamma_{N/quartz} = 9.3 \times 10^{-5} \exp(-3700/RT)$. The loss probability of the N atoms was also determined at 300 K on iron foils. The weak value obtained $\gamma_{N/iron} = 6.7 \times 10^{-4}$ is probably due to the nitriding of the metal. © 1999 Elsevier Science S.A. All rights reserved.

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1. Introduction

The control of post-discharge nitriding processes requires monitoring of the density of nitrogen atoms in the reactors, this species being responsible for the nitriding [1]. Nitrogen atoms are lost by either volume recombination by a three body reaction or by surface reaction [2,3]. In this latter case, the kinetics of recombination is strongly dependent on the nature of the surface [4]. It is also modified by the temperature [5]. The surface recombination of the nitrogen atoms has been investigated by many workers using a variety of experimental techniques [6,7]. Nevertheless, values of the loss probability γ_N given by the literature can be wide spread for a single material. For example on Pyrex, in the same experimental conditions, Yamashita et al. [8] got a value for $\gamma_{N/Pyrex}$ equal to 3.2×10^{-6} whereas $\gamma_{N/Pyrex}$ is 2×10^{-5} according to Ricard et al. [4] and 2×10^{-4} according to Gordiets et al. [9]. These differences, for a single material, in the same conditions of temperature and pressure, are due to the specific state of the surfaces, which are different in each case. Parameters like the roughness, the composition

and the crystallography of the surface, as well as the contamination by impurities can significantly affect the results [10–12]. A lot of work on Pyrex was carried out. Data on quartz is scarce. Since Evenson et al. [11], it is known that in pure nitrogen microwave discharges, the loss probability of atomic nitrogen on quartz walls increases from 7×10^{-6} – 5×10^{-4} when the total pressure increases from 400–4700 Pa.

The temperature dependence of the loss probability is usually given by an Arrhenius law. Little data is available on the temperature dependence of the loss probability of atoms on surfaces. For surface treatments, especially for nitriding, this dependence is a key parameter to control the process. A method of measurement of the dependence of the loss probability of atomic nitrogen on various surfaces is described herein. It is based on optical emission spectroscopy measurements in the Lewis–Rayleigh afterglow of an Ar-N₂ plasma, according to the principle already described by Ricard et al. at room temperature [4]. The influence of the temperature requires the development of a 2D hydrodynamic, thermal and kinetics model of the post-discharge. The modelling was applied, first with a quartz tube and secondly, the loss probability of N on iron was measured at room temperature.

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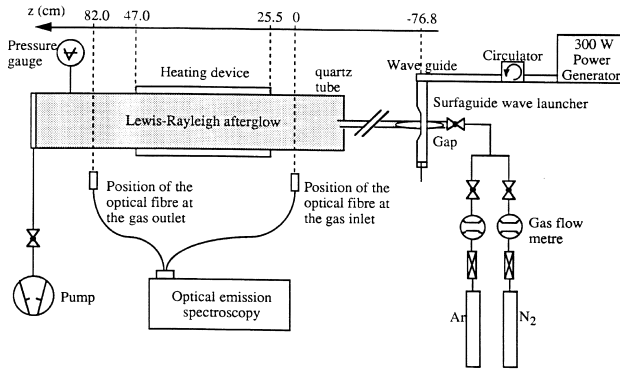


Fig. 1. Schematic representation of the experimental set-up.

2. Experimental devices

The experimental device used here is made of two cylindrical quartz tubes (Fig. 1). The first one, of 5 mm inner diameter, passes through a surfaguide wave launcher that creates an Ar–N₂ microwave discharge at 2.45 GHz. This first tube penetrates, at nearly 60 cm from the gap into a second quartz tube of 28 mm inner diameter where the spectroscopy measurements are carried out in the Lewis–Rayleigh afterglow. The distance of 60 cm is chosen so as to be in the late post-discharge. Indeed, in this zone, the intensity of the nitrogen first positive system (1st +) is proportional to the square of the nitrogen atom density [13].

To follow the temperature variation of the loss probability of nitrogen atoms at the walls, a heating device is applied around the second quartz tube (Fig. 1). A temperature profile can be settled by this means at the walls. When the reference temperature is in the range (300–823 K), the wall temperature evolves according to the law $T(z) = 300 + (T_r - 300) \exp(-(z - z_0)^2 / 0.022)$ where T_r is the reference temperature and z_0 is the distance in meters where the maximum temperature, equal to T_r , is obtained.

Two optical fibres are located at the height of the reactor axel, far enough upstream and downstream from the heating device to ensure a gas temperature around 300 K. They allowed to measure the intensities of the 1st + to calculate relative values of the N density. To obtain absolute values of the atomic nitrogen density from the intensities of the 1st +, titration of N atoms by NO, as described in [14] has to be carried out. The validity of this method was only demonstrated at room temperature. Titration is done at the position of the first optical fibre where $T = 300$ K. The total pressure is taken equal to 1500 Pa. The flow rates of argon and nitrogen are chosen, respectively, equal to 1000 and 50 sccm to compare our measurements to those available in [14]. A power of 20 W is delivered to the plasma. Under the experimental conditions described above, the mass fraction of atomic nitrogen measured by titration by NO at the position of the first optical fibre is equal to 2.46×10^{-4} , i.e. a molar concentration of 4.16×10^{-4} mol m⁻³ (or 2.5×10^{14} cm⁻³). This technique also offers the main advantage to

easily study different surfaces. The quartz tube itself was used as well as pure iron foils (25 μm thick and 21.5 cm long), located in the heated zone of the device. In this case, only measurements at room temperature were carried out. The temperature influence, which is more complicated because of diffusion, is in progress. Preparation of the surface before measurements is described in [14]. The surface roughness factor [9] was assumed to be equal to 1.

3. Description of the hydrodynamic model

3.1. Hypothesis and equations used

The hydrodynamic model is based on the solution of the coupled transport equations, corresponding to the continuity equations, the conservation of the momentum and the conservation of heat. The equations are solved in the steady state regime, assuming a continuous behaviour in the fluid [15].

Despite that the temperature gradients are important, Soret diffusion is herein ignored. No data are available on the thermal diffusion of N. However, we tried to estimate this coefficient by the following method. Using the value of the diffusion coefficient of N in N₂ available in [16], $D_{N/N_2} = 1.78 \times 10^5 T^{1.88} / P$, it is possible to determine the critical parameters of N from the equations of Wilke and Lee [17] giving the expression of the binary diffusion coefficient of a species from the Lennard–Jones potentials. With these values, it is possible to apply the Clark–Jones model [18] to estimate the thermal diffusion factor a_{12} and therefore the thermal diffusion coefficient D_T of N in N₂. Indeed, D_T is linked to D_{N/N_2} by

$$D_T = \omega_N \omega_{N_2} a_{12} D_{N/N_2} \quad (1)$$

where ρ is the density of the gas, ω_N and ω_{N_2} the mass fractions of N and N₂. By this method, the critical temperature obtained for N is $T_c = 170.5$ K with a collisional diameter $\sigma_N = 0.265$ nm. Therefore, one obtains a_{12} varying from 0.094 at $T = 300$ K to 0.145 at $T = 800$ K in N₂. Assuming $D_{N/N_2} = D_{N/Ar-4.76\%N_2}$, one has $|D_T| < 1.29 \times 10^9 T^{1.22} / P \text{ m}^2 \text{ s}^{-1}$ as $\omega_N < 2.46 \times 10^{-4}$. This weak value ($D_T / D_N \approx 10^{-5}$) is mainly due to the weak value of the mass fraction of N. It seems valid to neglect the Soret effect in our conditions.

Therefore the set of equations to solve here is

$$\nabla \cdot (\rho \vec{v} \omega_i - D_i \rho \vec{\nabla} \omega_i) = S_i \quad (2)$$

$$\nabla \cdot (\rho \vec{v} \vec{v}) + \nabla \cdot \vec{\tau} = -\vec{\nabla} P + \rho \vec{g} \quad (3)$$

$$\nabla \cdot (\rho \vec{v} C_p T - \lambda \vec{\nabla} T) = 0 \quad (4)$$

where ρ is the density of the gas, $\vec{v}$ is its velocity, ω_i is the mass fraction of species i and D_i is the binary diffusion coefficient in a Ar–4.76% N₂ mixture. $\vec{\tau}$ is the stress tensor,

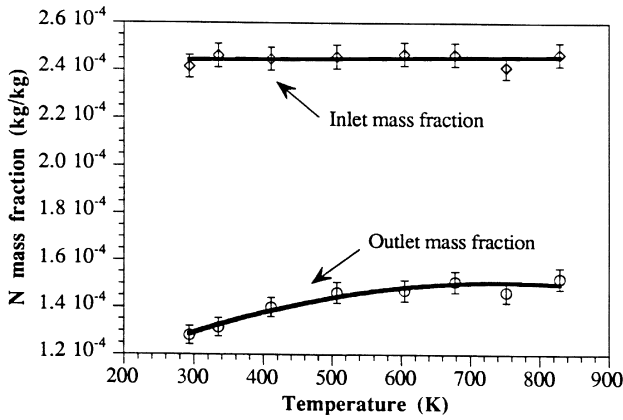


Fig. 2. Evolution of the atomic nitrogen concentration at the outlet of the device versus the temperature for a microwave power of 20 W. Walls are made of quartz only.

$\vec{g}$ is the acceleration due to gravity, T is the gas temperature, C_p is the specific heat at constant pressure and λ is the thermal conductivity. Transport data values are available in [15]. The pressure P is calculated from the ideal gas law. S_i is the source term in the balance equation of species i , i.e. the chemical reactions. For $i = N$, the term S_N^v represents the atomic nitrogen losses in the post-discharge zone due to chemical reactions. These losses are caused by the recombination of atomic nitrogen within the gas via the following mechanism



For the following reaction (a), $S_N^v = -2k[N]^2[M]$ with $k = 1.6 \times 10^3 \exp(T/273)^{-2} \text{ m}^6 \text{ mol}^{-2} \text{ s}^{-1}$ according to [19,20] and $M = [N_2]$ or $[M] = [\text{Ar}-N_2]$.

3.2. Boundary conditions

Atomic nitrogen is also consumed on the reactor walls by the recombination. The loss term S_N^s is treated as a boundary condition. In a first step, it is assumed that atomic nitrogen recombines at the surface by a first order Eley–Rideal

mechanism above 300 K on quartz. It can be found in [5] that this assumption is valid on Pyrex above 400 K. Therefore, one has



and $S_N^s = -k_{co}[N]$ where k_{co} is the rate constant for the disappearance of atomic nitrogen. At the walls, the rate constant is given by $k_{co} = \gamma_N v_{th}/D_H$ expressed in units of s^{-1} , where $D_H = 2R$ for a tube radius of R , v_{th} is the thermal velocity and γ_N is the loss coefficient [21]. The gas velocities were considered to be zero at the reactor walls. The gas temperature was taken equal to 300 K at the position of the first optical fibre. The equations of conservation of mass, momentum and heat were solved numerically using the implicit Phoenix finite volume calculation code (Phoenix 2.0, Concentration Heat and Mass Ltd., Bakery House, 40 High St., Wimbledon, London, SW19 5AU, England). Because of the symmetry of the installation, a 2D model was used with a cylindrical meshwork of 10×100 cells in r and z directions.

4. Results

In a first step, the variation of γ_N on the quartz tube versus temperature was studied. The emission of the 1st at the outlet of the device, coupled with NO titration, leads to a variation of the N atom mass fraction (ω_N) with temperature given in Fig. 2. It can be noticed that ω_N , measured at room temperature according to the principle already described, increases while the temperature of the heated zone increases. This is due to the fact that the three body recombination in volume (reaction (a)) is a decreasing function of temperature and that this decrease is higher than the rise of the rate of the surface recombination process.

Results of the calculations by the model are reproduced in Fig. 3. Values of calculated lost probability of N on quartz are very close to those given by Evenson et al. [11] at 1500 Pa: about 2.1×10^{-5} if one assumes an exponential dependence of γ with the total pressure (see [16]). The temperature dependence of γ_N on a quartz surface is an Arrhenius law. It can therefore be concluded that the recombination mechanism of N on quartz surfaces is the assumed, i.e. an Eley–Rideal mechanism. The activation energy is then estimated at $3700 \pm 500 \text{ J/mol}$. The pre-exponential factor is within the range: $\gamma_0 = 9.3 \pm 2.0 \times 10^{-5}$. The accuracy is calculated from several measurements.

The use of an iron foil leads to several difficulties due to the modification of the surface by nitrogen atoms. This can be seen in Fig. 4. The outlet mass fraction of nitrogen atoms becomes steady after 15 min. At this time, a nitride layer of several monolayers is probably formed [22].

A simple analytical calculation can be carried out when the temperature is constant over the whole distance in the post-discharge by solving the following equation

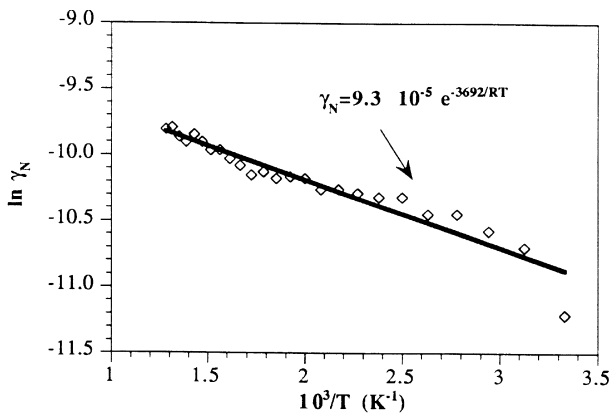


Fig. 3. Nitrogen atom loss probability calculated versus the temperature on a quartz surface.

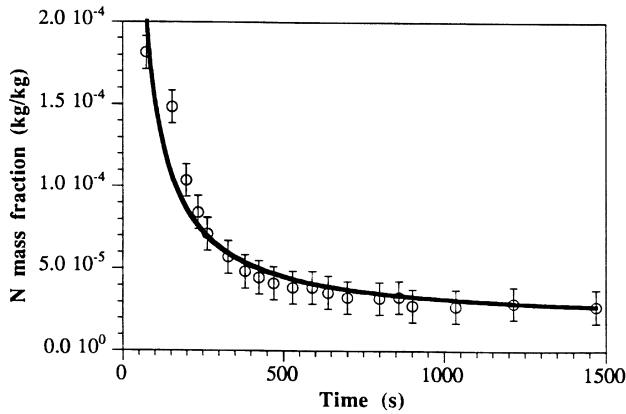


Fig. 4. Evolution of the atomic nitrogen concentration at the outlet of the device versus the time when an iron foil is introduced in the heated zone of the device.

$$\frac{dN}{dt} = -2k_v N^2 M - k_{\text{surf}} N \quad (5)$$

for the three zones detailed in Fig. 1 (a first zone from 0–25.5 cm where the surface is quartz, a second zone from 25.5–47 cm where the surface is iron and a third zone from 47–82 cm where the surface is quartz). $k_{\text{surf}} = k_{\text{quartz}}$ or k_{iron} . N concentration at the outlet is linked to inlet concentration by

$$\frac{1}{N_{\text{out}}} = -\alpha_{\text{quartz}} + \left\{ \alpha_{\text{quartz}} - \alpha_{\text{iron}} + \left(\alpha_{\text{quartz}} - \alpha_{\text{iron}} + \frac{\alpha_{\text{quartz}} N + 1}{N_{\text{in}}} e^{k_{\text{quartz}} t_1} \right) e^{k_{\text{iron}} t_2} \right\} e^{k_{\text{quartz}} t_3} \quad (6)$$

where $\alpha_{\text{quartz}} = 2k_v M / k_{\text{quartz}}$, $k_{\text{quartz}} = \gamma_{\text{N/quartz}} v_{\text{th}} / D_{\text{H}}$, t_i , $i = 1-3$ is the residence time of the N atoms in the inlet, heated and outlet parts of the device. The same notations are used for iron. This leads to $\gamma_{\text{N/iron}} = 6.7 \times 10^{-4}$. This value is weak when compared to 10^{-2} which is the most common value encountered in the literature for $\gamma_{\text{N/iron}}$. This is probably due to the fact that there is no surface sputtering in the post-discharge, whereas sputtering occurs on the cathode in the DC-discharges usually used to measure the loss probability and permits the maintaining of a metallic surface in contact with the gas phase. Further investigations based on Auger analyses will be carried out to study the influence of

the surface modifications on the loss probability of the nitrogen atoms.

5. Conclusion

The loss probability of nitrogen atoms on quartz was determined as a function of temperature for a power delivered to the plasma equal to 20 W. The loss mechanism at the walls was established to be an Eley–Rideal mechanism at a temperature ranging from 300–823 K. On iron, the loss probability of nitrogen atoms was measured at room temperature. Further investigations dealing with the influence of temperature are in progress. They require to take into account phenomena like the modification of the surface and diffusion.

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